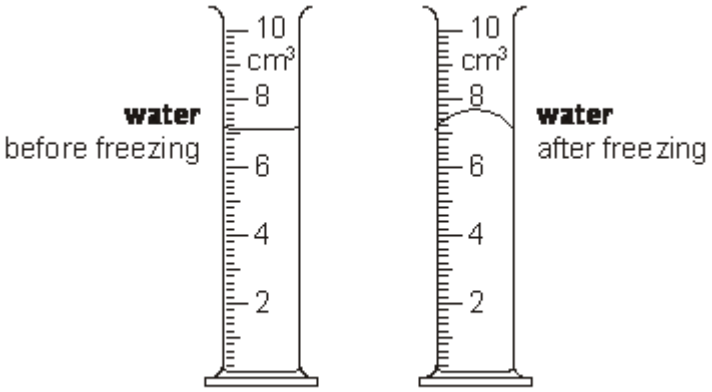
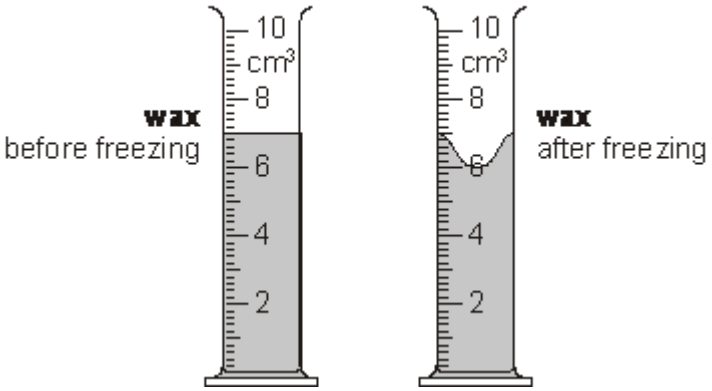
Q1. Meera poured 7 cm^3 of water into a measuring cylinder.

She poured 7 cm^3 of melted wax into another measuring cylinder.
She put both measuring cylinders into a freezer for 24 hours.





(a) Look at the measuring cylinders.
What happened to the volume of the water and the wax after freezing?

the volume of water .....

the volume of wax .....

1 mark

(b) The measuring cylinders were taken out of the freezer and left in a room at 20°C.

- Frozen water melts at 0°C.
- Wax melts at 53°C.

What would the physical state of each substance be at 20°C?

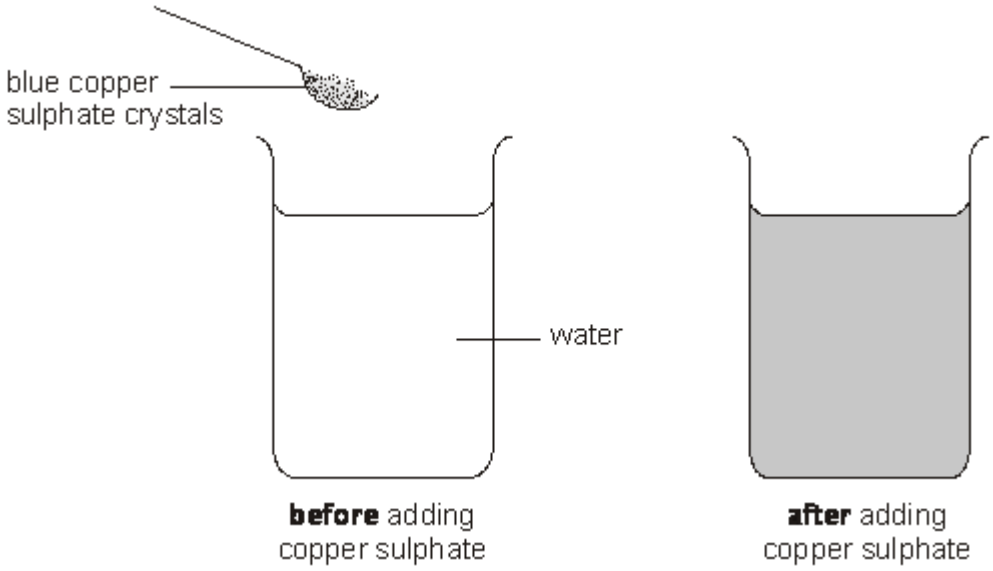
Choose from **gas** or **liquid** or **solid.**

water .....

wax .....

2 marks

(c) Meera added blue copper sulphate crystals to some water in a beaker.
The copper sulphate dissolved in the water.



(i) Give **one** way Meera could see that the copper sulphate had dissolved in the water.

.....

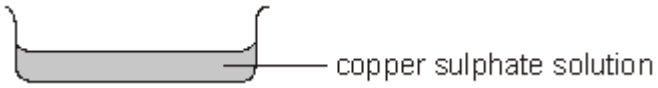
1 mark

(ii) Give **one** way that she could get the copper sulphate to dissolve more quickly.

.....

1 mark

(d) Meera poured some of the copper sulphate solution into a dish.
She left it in a warm room for a week.



A week later there was a blue solid but **no** liquid in the dish.

(i) What happened to the water in the copper sulphate solution?

.....

1 mark

(ii) What was the blue solid left in the dish?

.....

1 mark

maximum 7 marks

Mark schemes

Q1.

(a) • water: increased **or** went up

accept 'expanded'

*accept a reading greater than 7 (cm^3) and less than **or** equal to 8 (cm^3)*

• *wax*: decreased **or** went down

accept 'contracted'
*accept a reading lower than 7 (cm^3) and greater than **or** equal to 5.5 (cm^3)*

***both** answers are required for the mark*

1 (L3)

(b) • *water*: liquid

1 (L4)

• *wax*: solid

1 (L4)

(c) (i) any **one** from

• the liquid had turned blue

accept 'the colour changed'
*accept 'the water was blue **or** coloured **or** grey'*
'it formed a solution' is insufficient

• crystals **or** copper sulphate **or** solid could not be seen

accept 'the crystals disappeared'

• there was no solid left

1 (L3)

(ii) any **one** from

• stir it

*accept 'shake it' **or** 'mix it'*

• heat it

• use warm water

accept 'use hotter water'
accept 'add more water'
*do **not** accept 'put less in'*

1 (L3)

(d) (i) it evaporated

accept 'it went into the air'

1 (L4)

(ii) copper sulphate

accept 'crystals'
accept 'sulphate'

1 (L3)

[7]